

The Department of Petroleum and Gas Engineering operates under the Faculty of Engineering and Architecture. Established in the Fall of 2004 as the pioneering department of the newly formed engineering faculty, it was created to address the industrial need for managing Balochistan's extensive natural oil and gas reserves, such as the Sui gas fields. The four-year undergraduate program is fully accredited by the Pakistan Engineering Council and follows the Outcome-Based Assessment guidelines. The core focus of the department involves equipping students with scientific, mathematical, and computer-based skills for the exploration, drilling, appraisal, and optimal development of oil and gas assets. The mission of the program cultivates industrious engineers for the national and international petroleum and gas industry and its related fields in education, business, and government by providing technical knowledge, skills, and values through academic inquiry, teaching, and technical services. Graduates of the program are expected to achieve specific Program Educational Objectives within three to five years of graduation. These objectives include addressing complex problems in fields like renewable and non-renewable energy and sustainable development by utilizing professional, teamwork, communication, and managerial skills. Furthermore, graduates are expected to fulfill the requirements and necessary skills for higher academic studies and cutting-edge research, and to adopt a healthy and symbiotic environment with a strong commercial instinct, helping create new opportunities, especially in Balochistan.

For the Bachelor of Science in Petroleum and Gas Engineering program, the admission requirements state that an applicant's academic background must include an F.Sc Pre-Engineering from a recognized board or an equivalent with at least sixty percent marks, or a Diploma of Associate Engineer in a relevant field with at least sixty percent marks. Additionally, testing requires an entry test administered by NTS. The degree requirements mandate completing a total of 136 credit hours across 44 total courses, maintaining a minimum academic standing with a CGPA equal to or greater than 2.0, and successfully passing all 12 standard Program Learning Outcomes, which range from Engineering Knowledge and Problem Analysis to Ethics and Lifelong Learning. The curriculum plan details the program schema by semester, where credit hour structures denote theory plus lab hours equaling the total credit hours.

In the first semester, students take Fundamentals of Petroleum Engineering with no prerequisite for two and zero credit hours, Applied Geology with no prerequisite for two and one credit hours, Islamic Studies or Ethics with no prerequisite for two and zero credit hours, and Applied Mathematics-I with no prerequisite for three and zero credit hours. The first semester also includes Applied Physics or Functional English-I with no prerequisite for three and one or three and zero credit hours respectively, and Workshop Practice with no prerequisite for zero and two credit hours. The second semester consists of Communication Skills-II or Pakistan Studies with no prerequisites for three and zero or two and zero credit hours respectively. The third semester requires Applied Mathematics-II with a prerequisite of Applied Mathematics-I for three and zero credit hours, Applied Chemistry with no prerequisite for two and one credit hours, Engineering Drawing with no prerequisite for zero and two credit hours, and Introduction to Computer Programming with no prerequisite for two and one credit hours. The third semester continues with Technical Report Writing and Presentation Skills-III with no prerequisite for two and zero credit hours, Applied Mathematics-III with a prerequisite of Applied Mathematics-II for three and zero credit hours, Fluid Mechanics or Principles of Electrical Engineering with no prerequisites for two and one credit hours each, and Petroleum Geology and Geophysical Exploration or

Thermodynamics with no prerequisites for three and zero or two and one credit hours respectively.

The fourth semester encompasses Applied Statistics or Mechanics of Materials with no prerequisites for three and zero or two and one credit hours respectively, Drilling Engineering-I or Petrophysics with no prerequisites for three and one credit hours each, and Introduction to Sociology with no prerequisite for three and zero credit hours. The fifth semester involves Applied Numerical Methods with no prerequisite for two and one credit hours, Properties of Reservoir Fluids with no prerequisite for three and one credit hours, Drilling Engineering-II with a prerequisite of Drilling Engineering-I for three and one credit hours, and Environment and Safety Management or Process Control with no prerequisites for three and zero or two and one credit hours respectively. The sixth semester covers Reservoir Geomechanics with no prerequisite for two and zero credit hours, Well Logging with a prerequisite of Petrophysics for two and one credit hours, Reservoir Engineering-I or Petroleum Production Engineering-I with no prerequisites for three and one credit hours each, and Petroleum Refinery Engineering with no prerequisite for three and one credit hours. The seventh semester includes Final Year Project Part I with no prerequisite for zero and three credit hours, Project Planning and Management with no prerequisite for two and zero credit hours, Well Testing with no prerequisite for three and one credit hours, Petroleum Production Engineering-II with a prerequisite of Petroleum Production Engineering-I for three and one credit hours, and Reservoir Simulation with no prerequisite for three and one credit hours. The eighth semester concludes the program with Final Year Project Part II with no prerequisite for zero and three credit hours, Natural Gas Engineering or Economics with no prerequisites for two and one or two and zero credit hours respectively, Principles of Enhanced Oil Recovery with no prerequisite for three and one credit hours, and Reservoir Engineering with no prerequisite for three and one credit hours.